Hardware for drones

Battery - 1. weight, size , power distribution , power to the flight controller , power to other hardware , power to the motors,

Explore options of 2S, 3S, 6S

and plugs - distribution

Comm Providers

LoRA,

Private Base station - AmariSoft

WiFi - Mesh nodes - Drone to Drone // async transmission

// sync transmission - Vertiport scenario

file:///C:/Users/Shreeja%20Sridharan/Downloads/Aero_VersaWave_0_Brochure.pdf - Honeywell UAV Satcom terminal

https://web.archive.org/web/20240127063854/http://kosst.or.kr/data/Eyal.pdf

https://www.sigidwiki.com/images/d/d8/Inmarsats_Worldwide_Mobile-Satellite_Services_Today_and_Tomorrow.pdf

research related data from sensors can be saved offline on the ssd as well as be sent forward to ROS on our laptop (not GCS-QGroundControl imo)

Because It generates the mavlink as a packet with the serial inputs from other payload devices. On the comm terminal there runs a MAVLink router - its a forwarding system (which can also be run on Companion computer)

Then we use pixhawk only for the flight controller purpose and minimal standalone telemetry

Licensee point : Meshmerize

Advanced mesh designed for critical defense and
high-demand scenarios, ensuring maximum
resilience and performance.

The **Intel AC 7265** is a **Wi-Fi 5 (802.11ac)** adapter. It supports:

- **Wi-Fi Standards**: 802.11ac (Wi-Fi 5), 802.11a/b/g/n
- **Frequency Bands**: 2.4 GHz & 5 GHz
- **Max Speed**: Up to **867 Mbps**
- **Bluetooth Support**: Yes, Bluetooth 4.2

Using Intel AC 7265 with Jetson AGX Orin

NVIDIA Jetson AGX Orin does **not** have a built-in Wi-Fi module, so you need to manually install the **Intel AC 7265** card. Here's how:

1. Hardware Installation

- Jetson AGX Orin has **M.2 Key E** slot for Wi-Fi cards.
- The **Intel AC 7265 uses an M.2 Key E/A/E+ slot**, so it should be compatible.
- If using an **mPCIe version**, you may need an **M.2 to mPCIe adapter**.
- Attach the **antenna** for better Wi-Fi performance.

2. Install Wi-Fi Drivers

Jetson Orin runs **Ubuntu-based Linux (JetPack SDK)**. The Intel AC 7265 is supported by the **iwlwifi** driver, but you might need to install firmware.

Check if the card is detected:

```
lspci | grep -i wireless
```

or

```
dmesg | grep iwlwifi
```

Install Intel Wi-Fi firmware if missing:

```
sudo apt update
sudo apt install firmware-iwlwifi
```

Reload the driver:

```
sudo modprobe -r iwlwifi
sudo modprobe iwlwifi
```

3. Enable Wi-Fi

If the device is recognized, you can enable Wi-Fi from:

- **GUI**: Go to **Settings → Wi-Fi**
- **CLI** (if not listed in `ifconfig` or `ip a`):

  ```
  nmcli device wifi list
  nmcli device wifi connect <SSID> password <your-password>
  ```

4. Bluetooth Setup (Optional)

If you want to use Bluetooth:

```
sudo apt install bluez
sudo systemctl start bluetooth
bluetoothctl
```

Then scan and pair devices.

Potential Issues & Fixes

1. **Wi-Fi not detected?**
    - Check with `dmesg | grep iwlwifi`
    - Ensure `firmware-iwlwifi` is installed
    - Try `rfkill list` to see if it's blocked

- Run `sudo rfkill unblock all`

2. **Bluetooth not working?**
   - Ensure `bluez` is installed
   - Try `sudo systemctl restart bluetooth`

Would you like a recommendation for a better Wi-Fi 6E adapter for Orin? 🚀

The **AGX201-0NS** is a variant of the **Forge Carrier Board** designed by Connect Tech for the NVIDIA® Jetson AGX Orin™ module. This carrier board is tailored for commercial deployments, offering a compact footprint of 155mm x 125mm. ṆciteÖturn0search0Ộ̣Ợ

LTE Integration in EMEA Regions

For LTE (Long-Term Evolution) connectivity, the AGX201-0NS provides an **M.2 Key B 3042/3052 slot**. This slot is specifically designed to accommodate LTE or 5G modem modules and is equipped with a **USB 3.2 interface**. Additionally, it features a **Micro SIM (3FF) card slot** for SIM card insertion. ṆciteÖturn0search5Ộ̣Ợ

To ensure compatibility and optimal performance in the EMEA (Europe, Middle East, and Africa) regions, consider the following steps:

1. **Selecting the Appropriate LTE Module**:
   - **Regional Band Support**: Choose an LTE module that supports the frequency bands prevalent in the EMEA regions. This ensures robust connectivity and compliance with local telecommunications standards.Ợ
   - **Carrier Certification**: Opt for modules that are certified by major carriers within your target deployment areas. This facilitates smoother integration and potential troubleshooting with local networks.Ợ

2. **Installation and Configuration**:
   - **Hardware Installation**: Insert the selected LTE module into the M.2 Key B slot on the AGX201-0NS carrier board. Ensure that the module is securely seated and that the accompanying antennas are properly connected to guarantee optimal signal reception.Ợ
   - **SIM Card Insertion**: Place an active SIM card into the Micro SIM slot corresponding to your chosen carrier. Verify that the SIM card is active and

has an appropriate data plan for your application.Ợ

- **Driver and Firmware Setup**: Depending on the LTE module selected, you may need to install specific drivers or firmware. Refer to the module manufacturer's documentation for detailed installation instructions.Ợ

3. **Network Configuration**:

- **Default Route Settings**: When configuring the LTE interface as the primary network connection, it's essential to set the default route to prioritize the LTE network. This can be achieved using network management tools or command-line instructions. Be aware that some users have reported challenges in setting the LTE interface as the default route, with the system defaulting to onboard Ethernet interfaces. For instance, discussions on the NVIDIA Developer Forums highlight such issues and potential troubleshooting steps. ṆciteÖturn0search7ỘỢ
- **Fallback Mechanism**: Implement a fallback mechanism that reverts to Ethernet (eth0) if the LTE connection becomes unavailable. This ensures continuous network availability and minimizes potential downtime.Ợ

By meticulously selecting compatible LTE modules and configuring the network settings appropriately, the AGX201-0NS can be effectively utilized for LTE connectivity in EMEA regions, ensuring reliable and high-speed data transmission for your applications.Ợ

The **Intel AC9260** and **Intel AC7265** are both Wi-Fi and Bluetooth combo modules, but they have several key differences in terms of performance, features, and capabilities:

1. Wi-Fi Performance

Feature	**Intel AC9260**	**Intel AC7265**
Wi-Fi Standard	Wi-Fi 5 (802.11ac, Wave 2)	Wi-Fi 5 (802.11ac, Wave 1)
Max Speed	Up to **1.73 Gbps**	Up to **867 Mbps**
MU-MIMO	Yes (2×2)	No
Bands	Dual-band (2.4 GHz & 5 GHz)	Dual-band (2.4 GHz & 5 GHz)

- The **AC9260** supports **Wave 2** features like **MU-MIMO**, allowing multiple devices to communicate more efficiently with the router.
- It also has a higher **maximum speed (1.73 Gbps vs. 867 Mbps)**.

2. Bluetooth Support

Feature	**Intel AC9260**	**Intel AC7265**
Bluetooth Version	**5.0**	**4.2**

- The **AC9260** has **Bluetooth 5.0**, which provides better range, lower power consumption, and higher bandwidth compared to **Bluetooth 4.2** on the AC7265.

3. Interface & Compatibility

Feature	**Intel AC9260**	**Intel AC7265**
Interface	PCIe + USB	PCIe + USB
Form Factor	M.2 2230 / 1216	M.2 2230 / NGFF
Antenna Configuration	2×2	2×2

- Both modules use **PCIe + USB** for connectivity and have a **2×2 MIMO antenna** setup.
- The **AC9260** is mainly available in **M.2 2230 and 1216** form factors, while the **AC7265** comes in **M.2 2230 and NGFF**.

4. Power Consumption & Efficiency

- The **AC9260** is built on a **newer architecture**, making it more efficient in handling data and power.
- **MU-MIMO support** in AC9260 helps improve power efficiency when multiple devices are connected.

5. Overall Summary

- **Intel AC9260** is a better choice for faster Wi-Fi speeds, **MU-MIMO support**, and **Bluetooth 5.0**.

- **Intel AC7265** is an older model with **half the speed (867 Mbps vs. 1.73 Gbps)** and **Bluetooth 4.2**.

If you need **higher performance and future-proofing**, go with the **AC9260**. However, if you're using an older device that doesn't need **MU-MIMO or Bluetooth 5.0**, the **AC7265** may still be sufficient.

Hardware talks...

To dos for hardware

1. Learn the videoas Tushar posted
2. PID
3. RC - Different data links and purpose
4. LTE in the SIYI
5. How to get mavlink data from flight logs
6. Setup Mesh nodes - test bed
7. Fix LTE - Intel Nuc

https://github.com/Cosys-Lab/Cosys-AirSim

AirSim, ROS